Software for COMP 630 Databases

Overview

First and foremost, installing software is **not necessary** for COMP 630 Databases. The following software is available online via Vocareum. Instructions for use in Vocareum is in another document. However, you might choose to install it for offline use. Some benefits of installing the software locally:

- You have the software available for personal use.
- You have the software available for use after this course.
- You have the software available for use without an internet connection.

We will be installing

- Jupyter Notebook – a tool to integrate text, images, and computation within a single document
- PostgreSQL – a relational database management system
- MongoDB – a non-relational database management system
- Neo4J – a non-relational database management system
- Libraries to connect Jupyter to the DBMSes.

We will present one series of steps to install the software to be used exactly as shown in the videos. If you know what you are doing, there are other options that are not discussed here.

If you run into problems, experience says the simplest fix is to uninstall each of the programs and try again, following the instructions more carefully. However, you can ask your peers and staff for help.

This document does not describe how to use the software, since that is described within the course content.

Installing Jupyter Notebook

In a browser, go to www.anaconda.com/download/ . Anaconda is a collection of lots of applications and libraries, including Jupyter Notebook. Download and install the latest Python3-based version for your system. During installation, use the default options.

Potential problems

During installation, if you get a warning about your username containing a space, ignore it.

Installing PostgreSQL

Before starting, choose and remember a password for the PostgreSQL superuser account **postgres**.

In a browser, go to www.enterprisedb.com/products-services-training/pgdownload . Download and install the latest version for your system. During installation, use the default options, except for the following:

- You'll need to specify your chosen password for the **postgres** account.
- On the final screen, uncheck the box for running StackBuilder.

The installation process also starts the PostgreSQL server and sets it to restart every time you restart the computer.

Potential problems

If the installer crashes, try whitelisting the installer in your antivirus program.

In Windows, if you get the error "database cluster initialization failed", uninstall PostgreSQL. Reinstall, but choose a folder outside of "C:\Program Files" for the data file.

Creating a PostgreSQL account

With the online service we're using, you'll automatically get a PostgreSQL user account (i.e., an account other than the all-powerful **postgres**). But here, you'll need to create one yourself. Choose and remember a user account name, user database name, and password. The examples provided during the course use 'riceuser', 'ricedb', and 'ricepassword', respectively. For better security, choose your own password.

Run the Postgres "shell":

- Windows: Start / PostgreSQL / SQL Shell
- Mac: Applications / PostgreSQL / SQL Shell

In this shell program, enter the following two commands, substituting the chosen values for the italics:

- CREATE USER *username* PASSWORD *'userpassword'*;
- CREATE DATABASE *userdatabase* OWNER *username*;

Installing MongoDB

In a browser, go to www.mongodb.com/download-center/community . Download and install the latest version for your system. During installation, use the default options.

The installation process also starts the MongoDB server and sets it to restart every time you restart the computer.

Installing Neo4j

In a browser, go to https://neo4j.com/download/ . Download and install the latest version for your system. During installation, use the default options.

Neo4j takes a lazy approach and doesn't actually do anything until necessary. That holds for installation as well. To complete installation, run Neo4j Desktop. During installation, use the default options, but note the folder used for data. This first time, you will need to register the software. Then select Add Graph, Create Local Graph, pick a password for this graph, and Run.

The installation process does *not* start the Neo4j server.

Changing Neo4j server password

By default, the Neo4j server userid and password are 'neo4j' and 'neo4j'. To change the password, you first need to install Java from www.java.com/download/ .

Start Anaconda's command line program:

- Windows: Start / Anaconda3 / Anaconda Prompt
- Mac: Applications / Anaconda Prompt

Type the following commands, substituting Neo4j's data folder which you noted previously:

- cd *neo4j_data_path*
- cd neo4jDatabases/database*/installation*/bin
- neo4j-admin set-initial-password ricepassword

Running database servers

In order to use the database servers from Jupyter Notebook, the server processes must be running. For PostgreSQL and MongoDB, the installation process starts the server process and sets it to restart every time you restart the computer. For Neo4j, you'll need to start the server process yourself, which is slightly awkward to do. The simplest way to do this is to follow these steps:

1. Run Neo4j Desktop
2. If there is no Graph yet defined, define one via Add A Graph, Create Graph Locally, enter a password, and Create.
3. For any defined Graph, use Start.

We're not actually going to use that graph, but these steps force the server to be ready..

Tying Jupyter Notebook to the database servers

Run Anaconda Navigator:

- Windows: Start / Anaconda3 / Anaconda Navigator
- Mac: Applications / Anaconda Navigator

On the left, select Environments. In the pull-down menu, select Uninstalled.

Select the following libraries to install:

- psycopg2
- pymongo

Select Apply. In the popup window, select Apply.

Start Anaconda's command line program:

- Windows: Start / Anaconda3 / Anaconda Prompt
- Mac: Applications / Anaconda Prompt

Enter each of the following two lines to this program, installing more libraries:

- pip install ipython-sql
- pip install ipython-cypher

Adding better formatting to Jupyter

By default, Jupyter highlights all code as if it is Python. We'd like it to highlight SQL and Cypher syntax appropriately, so that it is more readable. Additionally, it is helpful to turn on line numbering by default. These two formatting changes will help you, but are **not** necessary.

Let's first check if the appropriate folder exists yet.

- Windows: In Windows Explorer, go to the folder C:/Users/*youruserid*/ and look for the folder .jupyter .
- Mac: In Finder, select Go / Home. Look for the folder .jupyter .

If that folder doesn't exist, we can create it. Start Anaconda's command line:

- Windows: Start / Anaconda3 / Anaconda Prompt
- Mac: Applications / Anaconda Prompt

and then type in the following command: jupyter notebook --generate-config

Create a folder called "custom" within the .jupyter folder. We provide files custom.js and custom.json in the Canvas Modules. Download and save these files into the .jupyter/custom folder. The next time you run Jupyter Notebook, it should format SQL and Cypher code nicely.

Potential problems

If your SQL code isn't nicely formatted, exit Jupyter Notebook, clear your web browser cache, and then restart Jupyter Notebook.